Term paper – ECON4170

Introduction
I am aiming to investigate global and regional Covid-19 virus testing trends from 2020 to 2024 using data from "Our World in Data." I will look into positive testing data, death cases, and vaccination rate in the world and to see if there are correlations between GDP per capita and positive test, and demography indicators.

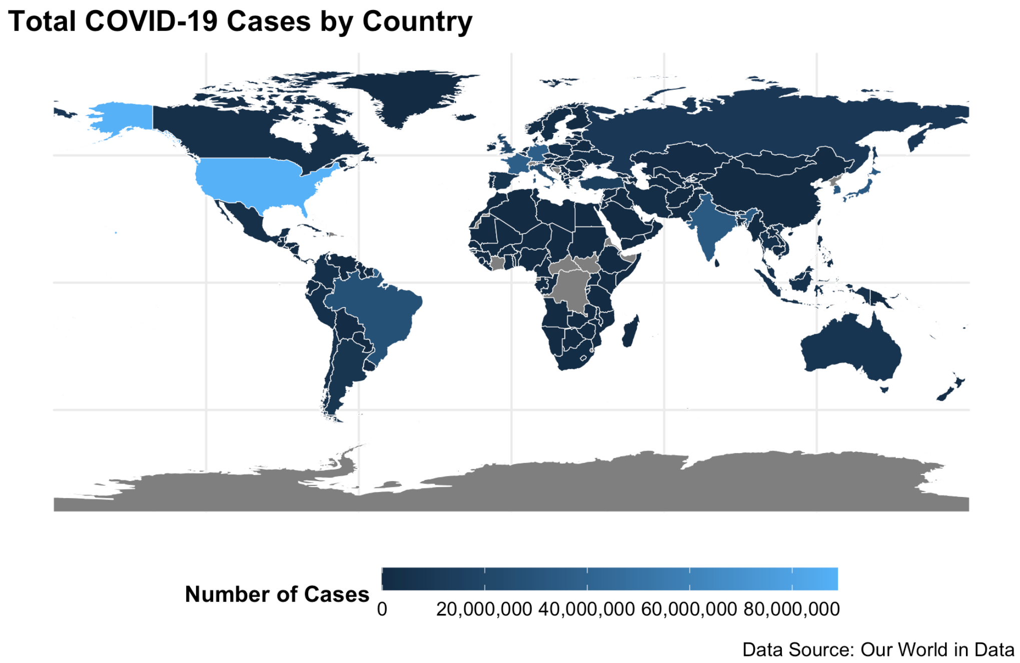
In this project, I will focus on gathering and wrangling data, as well as visualizing the findings in an aesthetic and representative way. These are the two focus areas of the project paper.

The testing data I will base my findings in, contains large testing observations from around the world. This creates a great foundation to analyses the global trends and I believe can also reflects the different policy makers decisions during these times.

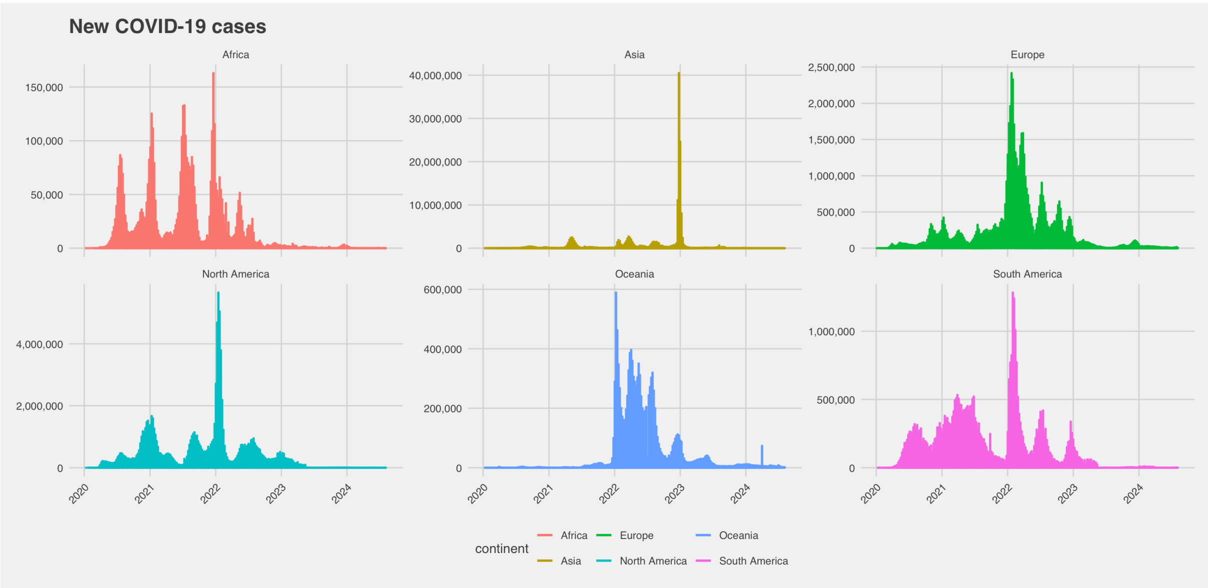
Main part:
Visualizations
For this part I have decided to use the packages ggplot2 and ggthemes to visualise the data. The FiveThirtyEight theme mimics the look of the graphics used on the FiveThirtyEight website, a well-known statistics and analytics platform. This style gives the charts a clean and professional look with a focus on clarity and readability.

The World Health Organization (WHO) declared coronavirus disease 2019 (COVID-19) an international health emergency on January 30, 2020, and later as a pandemic on March 11, 2020.

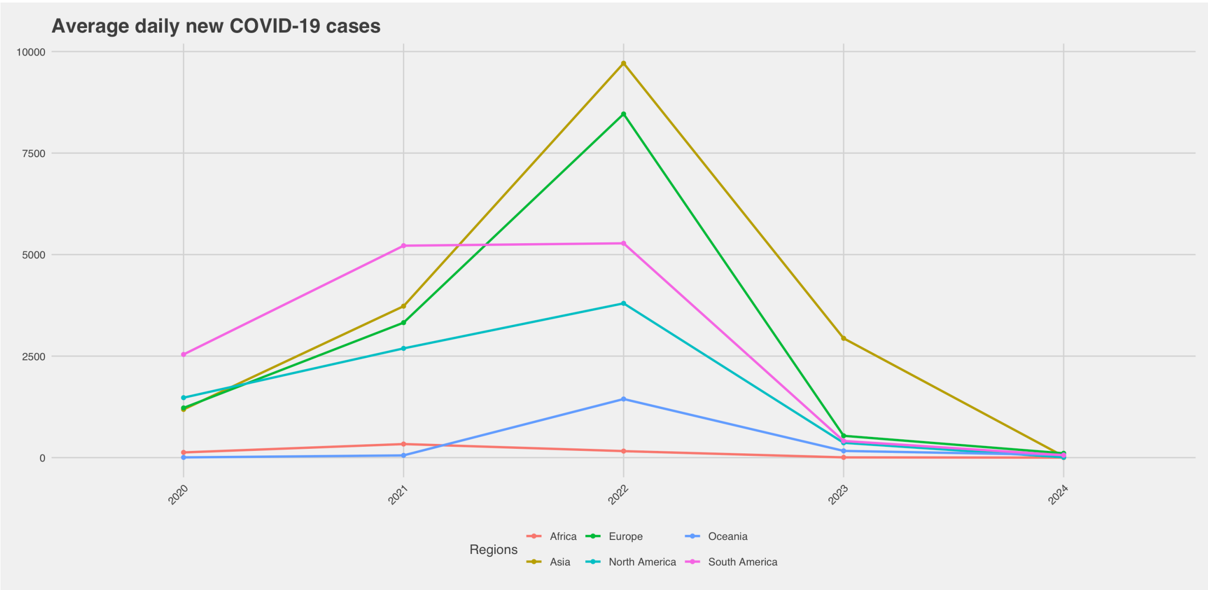
Let's start by looking at new covid cases in different regions in the world. The map below visualizes this data using a gradient, where the intensity of colour represents the number of new cases. From the map, we can observe that the highest concentrations of cases are found in the USA, parts of Europe, and India. These regions stand out due to their lighter colours, indicating a higher number of reported cases compared to other areas.



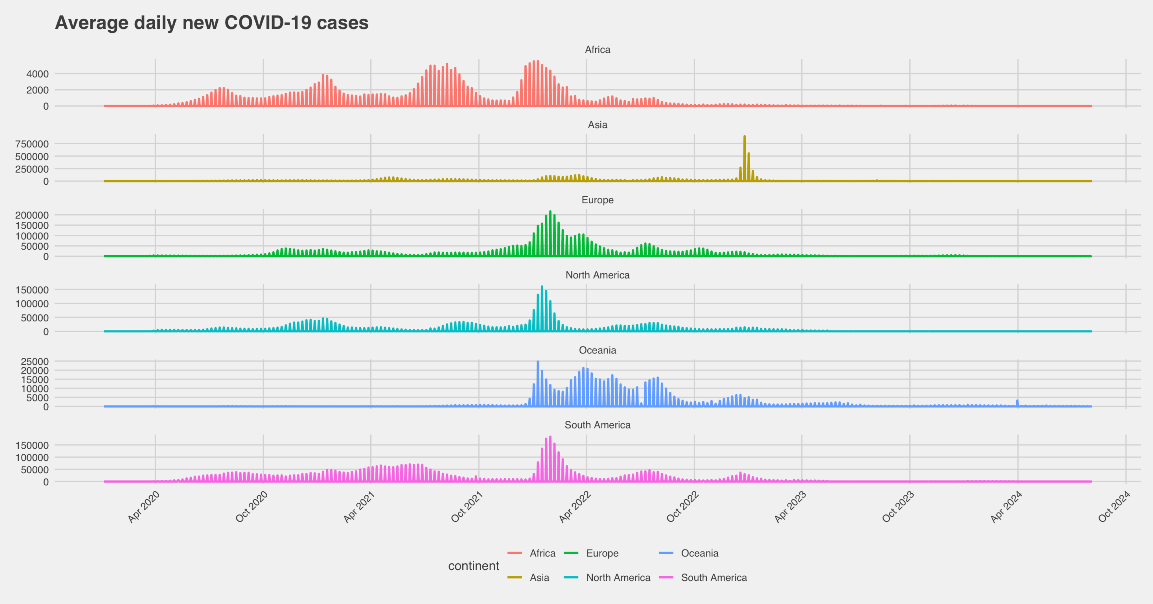
The graph below shows that the patterns of confirmed new cases of COVID-19, which means a person has tested positive for the virus, over the last four years from 2020 to 2024. We can observe that most regions experienced their peak in 2022, while Asia saw its peak in 2023. It is also important to mention that I have used a separate scale on the y-axis to provide a clearer picture of the trends for each area, without the large numbers from China 'hiding' developments in other regions."



Further investigation into the average number of new COVID-19 cases reveals some noteworthy trends. During the peak of the pandemic, Asia recorded an average of nearly 10,000 new cases per day, while Europe reached an average of 8,500 new cases daily. These figures highlight the intensity of the pandemic in these regions at their respective peaks. However, it is important to note that by 2024, all regions have shown a significant flattening of the curve, indicating reduction in the number of new cases across the globe and is under control. This trend suggests that, while some regions experienced more severe outbreaks than others, the global situation has improved overall from 2023 to today.

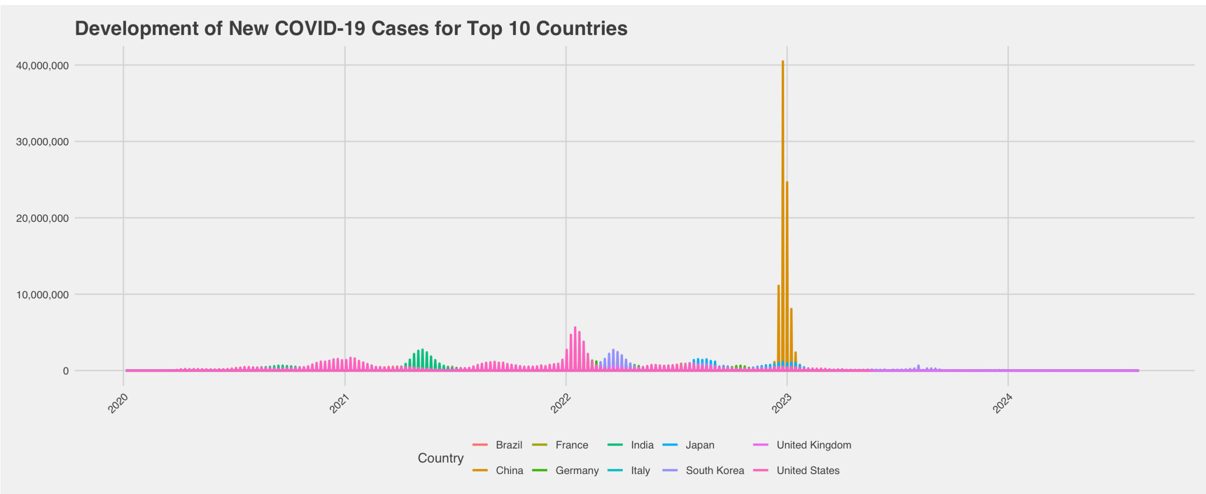


Here, we can clearly observe that global trends have flattened across all regions. As previously mentioned, a separate scale has been used on the y-axis to provide a clearer depiction of the trends in each area. While Africa experienced relatively moderate COVID-19 waves, Asia, for example, exhibited a sharp peak in 2023, highlighting the regional differences in the progression of the pandemic.



Furthermore, I wanted to examine the top 10 countries most affected by COVID-19 in terms of new cases, which include Brazil, China, France, Germany, India, Japan, South Korea, the UK, and the USA. These nations represent a significant portion of the global impact of the pandemic and provide insight into the regional disparities in the spread of the virus.

One surprising observation was that, despite COVID-19 originating in China, the number of new cases there remained relatively lower compared to other regions until its peak in 2023. This could be attributed to China's early and aggressive containment measures, such as strict lockdowns and mass testing campaigns, which may have initially limited the virus's spread. However, as restrictions eased or variants emerged, case numbers eventually surged, aligning with global trends but even higher. Before China's peak in 2023, the UK experienced the highest impact, reaching its own peak in 2022.



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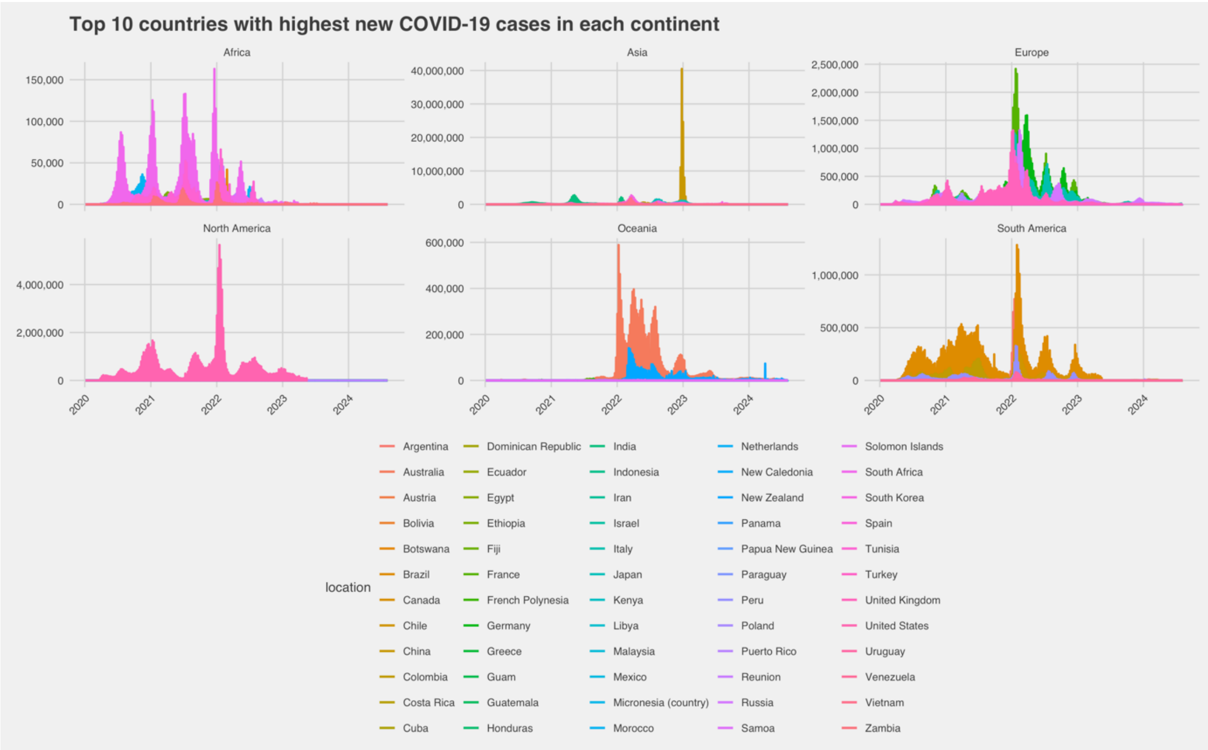
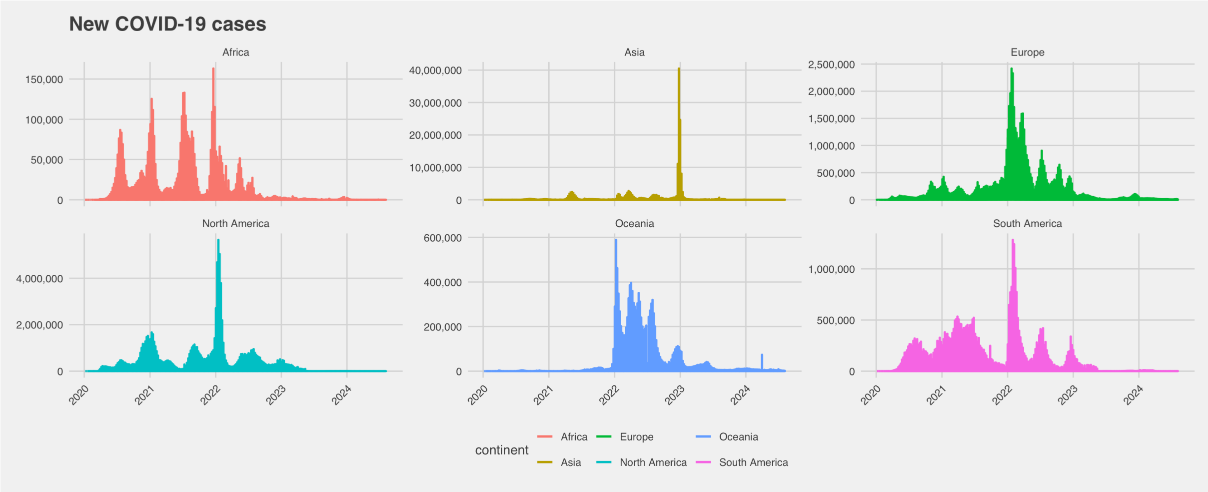
Upon further examination, I chose to analyse the top 10 countries within each region. The graph shows the ten countries with the highest number of new COVID-19 cases in each of the six continents: North America, South America, Europe, Africa, Asia, and Oceania. Each graph is color-coded to represent a continent, with the y-axis showing the number of cases while the x-axis shows the countries.



When examining total confirmed COVID-19 cases, the data highlights the vast differences in the pandemic's spread across continents. Starting with the regions reporting the lowest totals, Africa recorded just over 150,000 cases. Oceania followed with approximately 600,000 cases, reflecting its relatively smaller population and strict border controls in several countries. South America experienced over 1,000,000 confirmed cases, showcasing the significant impact of the pandemic on this densely populated region.

Moving to regions with higher case counts, Europe reported a staggering 2,500,000 cases, emphasizing the scale of the outbreaks in countries like France, Germany, and the UK. North America surpassed 4,000,000 cases, with the United States contributing the majority due to its population size and testing rates.

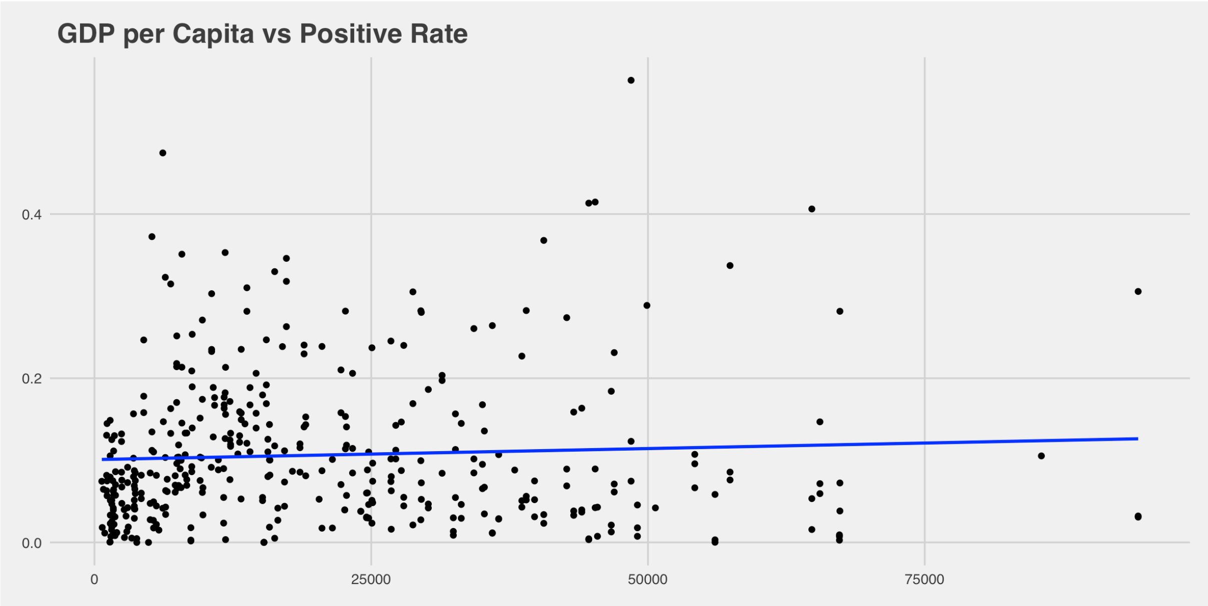
However, the largest share of total confirmed cases by far was recorded in Asia, with over 40,000,000 cases at its peak. This immense number reflects Asia's vast population, the concentrated outbreaks in countries like India and China, and the later surges in regions such as Southeast Asia. These figures underscore the varied dynamics of the pandemic's progression, shaped by population density and public health policies.



Examining correlations, I sought to explore the relationship between GDP per capita and the positivity rate. In this context, the positivity rate is defined as the proportion of a country's population that tested positive for COVID-19. Each dot on the scatterplot represents a country, providing a clear visual representation of the data.

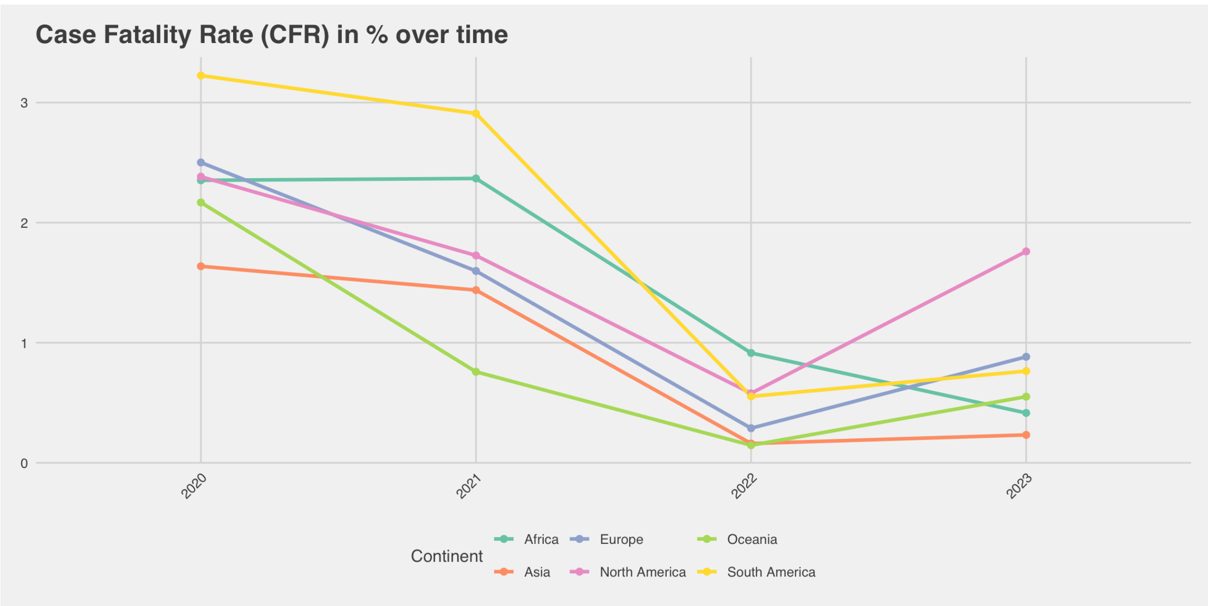
From the graph, we observe a slight positive correlation between GDP per capita and positivity rate. This means that countries with higher GDP per capita tend to have slightly higher positivity rates. While the correlation is not strong, it raises intriguing questions about the potential factors driving this trend.

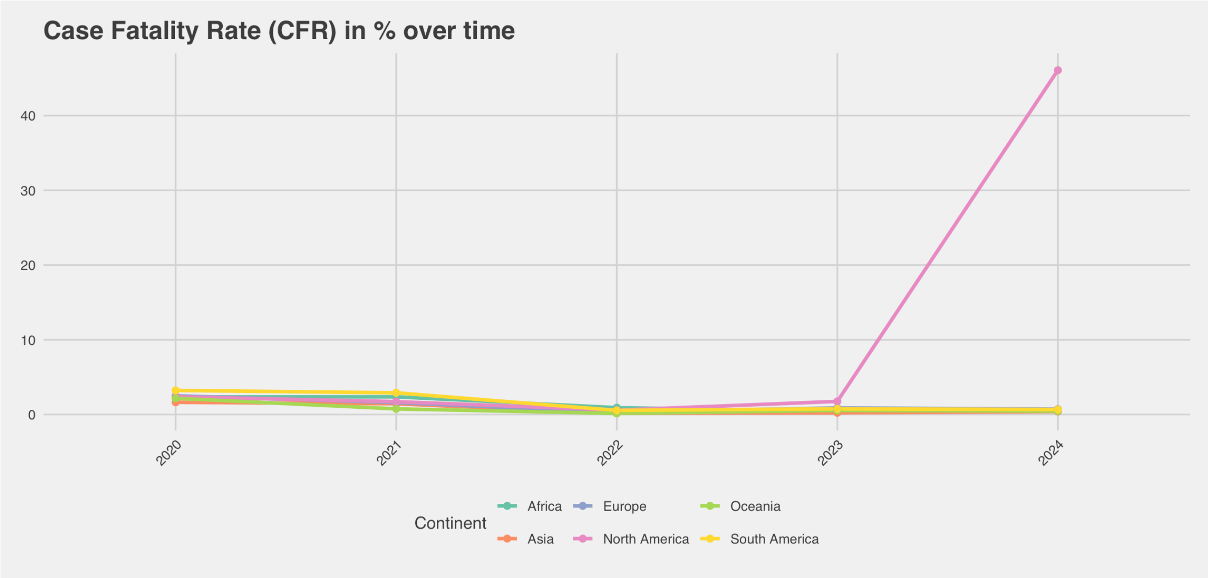
For instance, wealthier nations often have more extensive testing capabilities, which may result in higher reported positivity rates. They may also have more mobile populations, leading to greater exposure to the virus. Conversely, in lower-income countries, limited testing infrastructure and underreporting might mask the true positivity rate, potentially skewing the relationship.



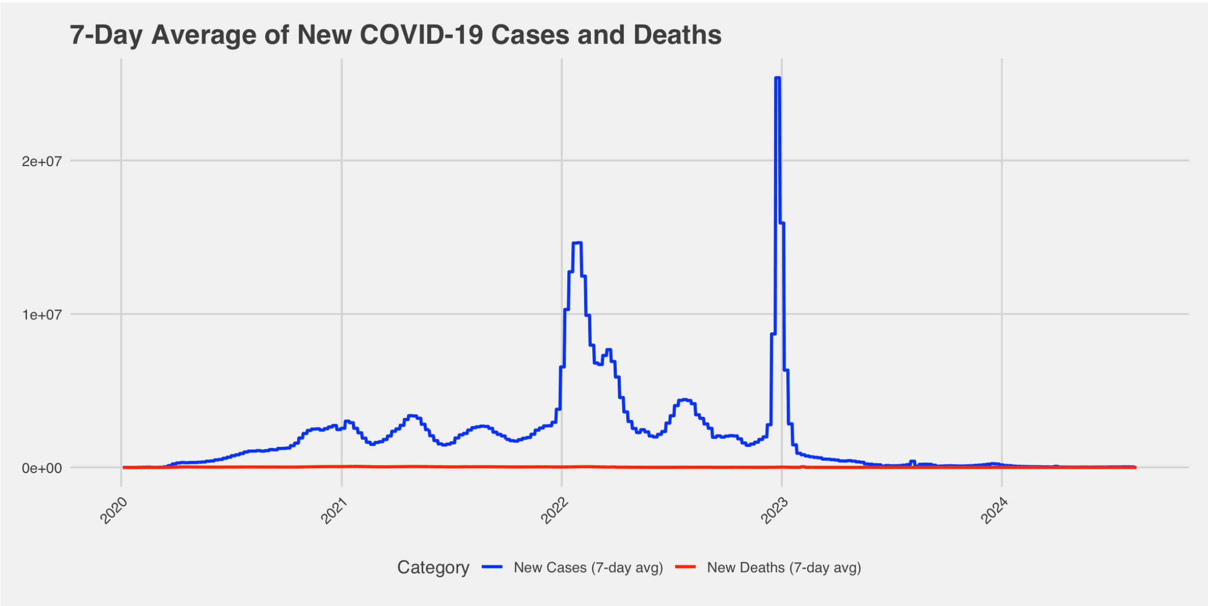
Case fatality rate:

When examining case fatality rates (CFR) from 2020 to 2023, an inverse trend emerges compared to the trajectory of new COVID-19 cases. While the number of new cases rose dramatically over time, particularly during the peaks in 2022 and 2023, the CFR—a measure of the proportion of deaths relative to confirmed cases—remained relatively stable. This stability in the absolute number of fatalities, coupled with the exponential growth in cases, resulted in a declining CFR over time. By bringing 2024, the data doesn`t explain the data as good.

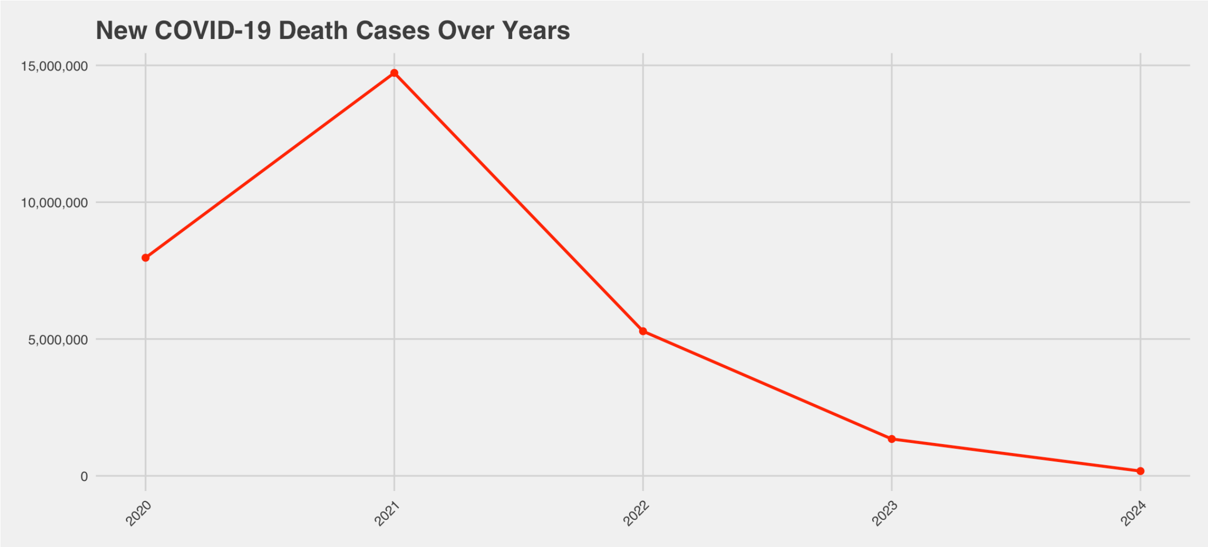


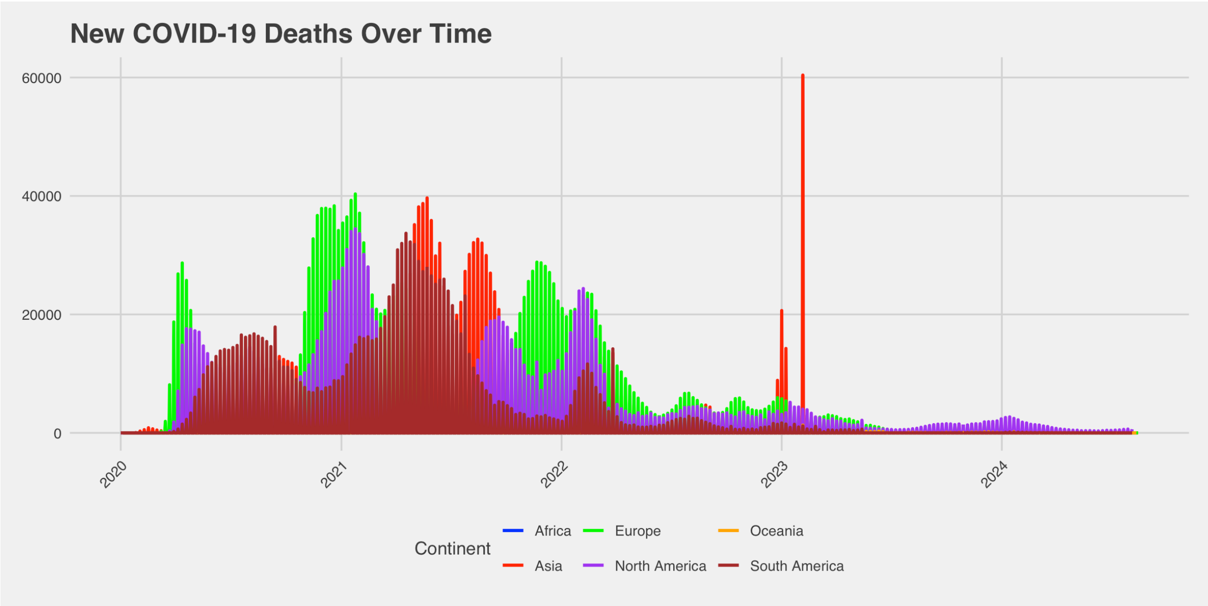


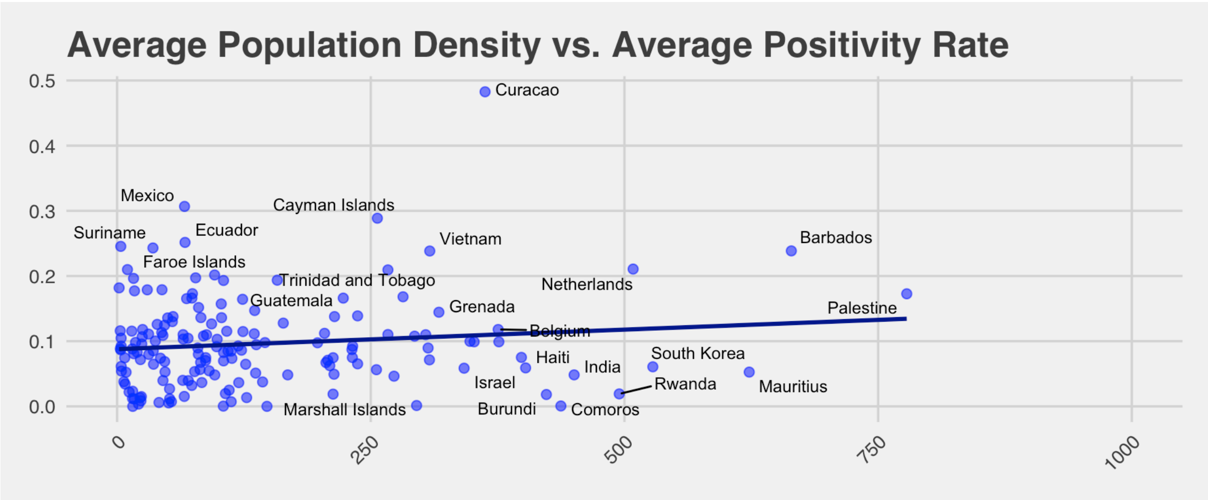
Deaths over the years:



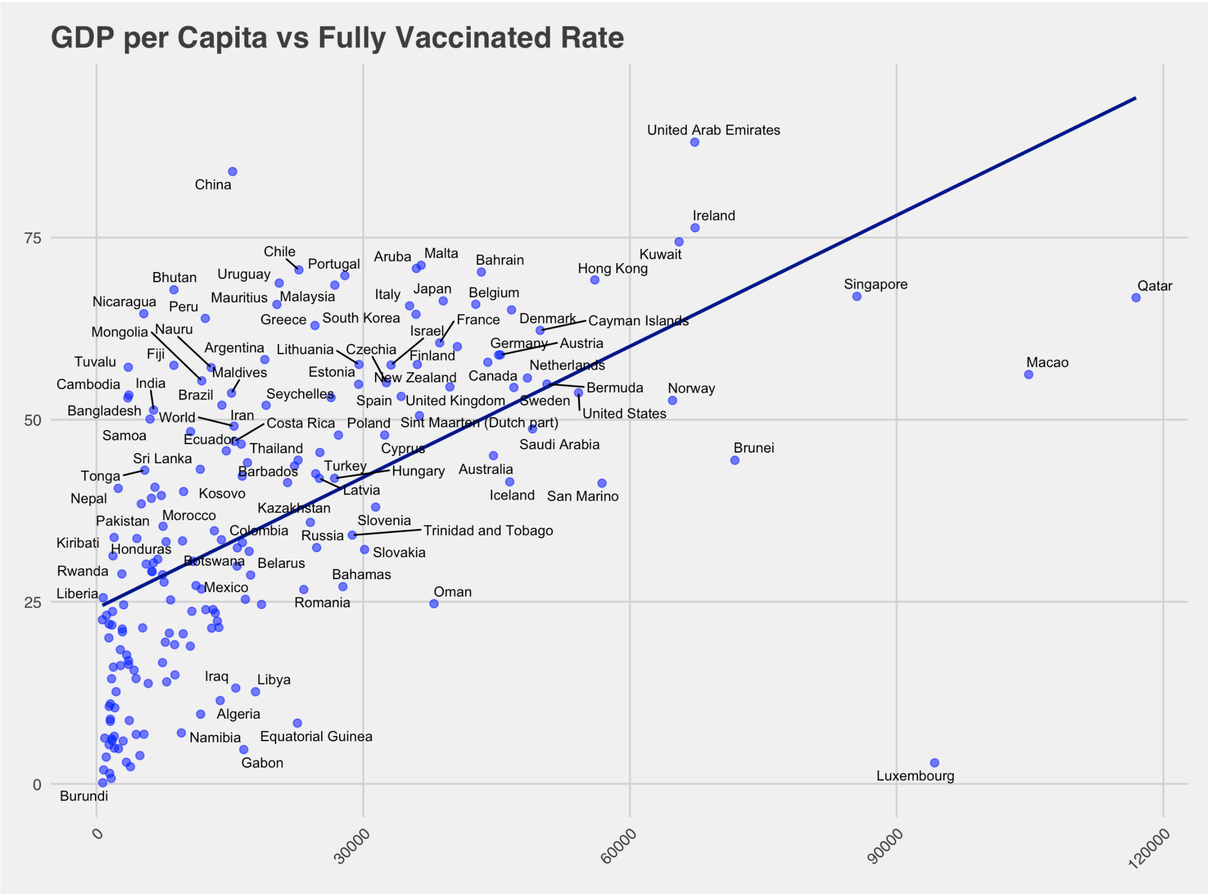
Examining COVID-19 death cases worldwide over the past four years reveals that the highest peak occurred in 2021. This year saw a significant increase in deaths, highlighting the cruelty of the pandemic during that period and how vaccines and other tools helped decreasing the trend.


New COVID-19 Death Cases Over Years
15,000,000
10,000,000
5,000,000
0
2020
2021
2022
2023
2024


New COVID-19 Deaths Over Time
60000
40000
20000
0
2020
2021
2022
2023
2024
Continent
Africa
Europe
Oceania
Asia
North America
South America


Average Population Density vs. Average Positivity Rate
0.5
0.4
0.3
0.2
0.1
0.0
0
250
500
750
1000
Curacao
Mexico
Cayman Islands
Suriname
Ecuador
Vietnam
Barbados
Faroe Islands
Trinidad and Tobago
Netherlands
Guatemala
Grenada
Palestine
Belgium
South Korea
Haiti
India
Israel
Rwanda
Mauritius
Marshall Islands
Burundi
Comoros

The graph under presents a strong positive correlation between GDP per capita and the fully vaccinated rate. As GDP per capita increases, the vaccination rate tends to rise as well, suggesting that wealthier countries have been more successful and more resources in vaccinating their populations.



Conclusion:

The findings reveal fascinating global trends in COVID-19 testing data, shedding light on how the pandemic unfolded across different regions and countries. This analysis underscores the significant role of data in understanding and responding to global health crises and how policy makers reflects the data. By visualizing trends, correlations, and regional differences, we gain a deeper understanding of how testing efforts varied and how they related to broader economic and demographic factors.

Additionally, this study highlights the power and versatility of R as a tool for working with large, complex datasets containing numerous observations and variables. R's robust libraries and data visualization capabilities make it particularly well-suited for handling multifaceted analyses like these. Whether exploring trends over time, generating maps to visualize geographical distributions, or creating scatterplots to examine correlations, R enables researchers to extract meaningful insights efficiently and effectively.

Sources/ references

https://ourworldindata.org/coronavirus-testing

Link to R code source:

https://github.com/lalitathorbjornsen/EXAM